

STANDUP 2 Study Protocol

Evaluation of Supracardiac Venous Stenting on Orthostatic Intolerance and Orthostatic Hypotension

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Background

Nearly a century ago, Bradbury and Eggleston provided the first documented description of postural orthostatic hypotension [1]. It is now widely recognized and is thought to affect approximately 16% to 30% of adults aged 65 years or older [2]. Orthostatic hypotension (OH), defined as a reduction of systolic blood pressure of at least 20 mmHg or diastolic blood pressure of at least 10 mmHg within 3 minutes of standing[3], a form of orthostatic intolerance (OI), which is characterized by the inability to tolerate the upright posture due to symptoms such as dizziness or syncope that are relieved by recumbency[3]. is associated with a significantly increased risk of all-cause mortality, incident coronary heart disease, heart failure, and stroke [4]. It is also associated with an increased risk of atrial fibrillation [5], venous thromboembolism [6], and chronic kidney disease [7]. Current pharmacological treatments such as midodrine, fludrocortisone, droxidopa, and pyridostigmine are suboptimal in terms of symptom relief and often limited by serious side effects [8], including severe supine hypertension [9], adverse urological effects [10], hypokalemia, metabolic alkalosis [11], congestive heart failure [12], headaches, nausea, palpitations [13], fatigue, and muscle weakness [14].

Systemic arterial pressure is maintained via a complex orchestration of physical, chemical, and biological processes [15]. We propose that supracardiac venous outflow impairment disrupts this regulation and contributes to orthostatic hypotension and orthostatic intolerance (OI). The scientific rationale for this involves several interconnected mechanisms:

Baroreflex and Metabolite Clearance

The body's primary defense against OH is the baroreflex mechanism. Baroreceptors in the carotid sinus and aortic arch detect blood pressure changes, transmitting information to the brainstem via the glossopharyngeal and vagus nerves. The nucleus tractus solitarius process this input, and the caudal and rostral ventrolateral medulla modulate sympathetic outflow to adjust heart rate and vascular tone [16]. Efficiency depends on proper metabolite clearance, governed by the Fick principle [17], where metabolite removal rate is proportional to the arterial-venous concentration difference and organ blood flow. When venous outflow is impaired, reduced clearance may delay the removal of metabolites from autonomic control centers. Experimental models have identified imidazoleacetic acid ribotide within vestibulo-sympathetic pathway neurons, where it modulates autonomic signaling [18]. Similarly, glutamate, known to accumulate during cerebral hypoperfusion, and extracellular potassium, elevated in ischemic conditions, can suppress sympathetic activity and blunt baroreflex sensitivity. Such disturbances provide a plausible mechanism by which cerebral venous congestion may contribute to autonomic dysregulation and orthostasis.

Venous Return and Cardiac Output Dynamics

Venous return is a critical determinant of cardiac output, governed by the pressure gradient between the mean systemic filling pressure (P_{MS}) and right atrial pressure (P_{RA}), and inversely proportional to the resistance to venous return (R_V), such that

$$VR = \frac{P_{MS} - P_{RA}}{R_V}$$

[19]. In the upright position, gravity causes pooling of more than 500 ml of blood in the venous capacitance vessels and splanchnic circulation, leading to a decrease in venous return and a drop in cardiac output of nearly 20% [8]. This produces a remarkable decrease in mean arterial pressure at the level of the head [20]. Cerebral autoregulation is less able to maintain stable perfusion during sudden hypotension compared to sudden hypertension [20]. Under normal circumstances, this is counteracted by multiple mechanisms for a smooth, asymptomatic postural transition. Venous valvular stenoses impede venous return proportionally to the number and degree of stenosed valves, consequently decreasing cardiac output.

Vestibulo-Sympathetic Reflex and Sympathetic Anticipation Failure

The vestibulo-sympathetic reflex integrates vestibular inputs from the inner ear with sympathetic nervous system activity to anticipate and counteract postural changes, such as standing. In OH patients with supracardiac venous outflow obstruction, venous congestion leads to poor metabolite clearance in key sympathetic pathways, including the rostral ventrolateral medulla and sympathetic preganglionic neurons. This impairs anticipatory sympathetic activation prior to standing, resulting in delayed or insufficient vasoconstriction and heart rate adjustments. Consequently, patients experience exaggerated drops in blood pressure upon postural change. Evidence from microgravity studies provides physiological support for this hypothesis. Prolonged exposure to weightlessness results in vestibular deconditioning, reflected by reduced otolith-driven reflexes and diminished sympathetic coupling[21]. Astronauts frequently exhibit orthostatic intolerance upon return to normal gravity, attributed to altered vestibular integration and blunted sympathetic response to postural changes[22]. Similarly, chronic cerebral venous congestion due to internal jugular vein narrowing may interfere with these same regulatory circuits, contributing to impaired sympathetic anticipation and orthostatic hypotension.

Static Venous Tone

We suspect that venous tone in OH patients is somewhat static, failing to dynamically adjust to postural demands. This manifests as excessive venous constriction in the supine position (contributing to supine hypertension) and inadequate constriction while standing (exacerbating hypotension) [20]. Venous outflow obstruction may perpetuate this dysregulation by maintaining chronic congestion, preventing adaptive venoconstriction. Improving venous outflow via angioplasty or stenting may help restore dynamic venous tone, potentially mitigating both supine hypertension and standing hypotension [19].

Glymphatic Clearance, Venous Engorgement, and the Fick Principle

A key underpinning mechanism contributing to the pathophysiology of orthostatic hypotension in the context of venous outflow obstruction is the disruption of the glymphatic system, a brain-wide perivascular network that clears metabolic waste. The glymphatic system, which is most active during sleep, relies on the active bulk flow of cerebrospinal fluid (CSF) into the brain along periarterial spaces, followed by the outflow of interstitial fluid and metabolic waste products along perivenous spaces. Glymphatic clearance is crucial for removing neurotoxic proteins and metabolites, including those that influence autonomic function. Cerebral venous outflow obstruction leads to increased intracranial pressure and venous congestion, which directly impacts the perivenous outflow component of the glymphatic system [19]. This venous engorgement can constrict the perivenous spaces, impeding the drainage of interstitial fluid. According to the Fick principle, the rate of metabolite removal is directly proportional to the blood flow [17]. When venous outflow is compromised, blood flow through the brain's microcirculation is slowed, reducing the efficiency of metabolite clearance from critical brainstem centers like the rostral ventrolateral medulla, which regulates sympathetic outflow [23]. This accumulation of metabolites can impair neuronal function, leading to a blunted or delayed sympathetic response to postural changes, a phenomenon we refer to as sympathetic anticipation failure. Therefore, supracardiac venous angioplasty or stenting, by improving venous outflow, may restore proper glymphatic clearance and metabolite exchange [19], thereby re-sensitizing autonomic centers and improving orthostatic tolerance.

Although therapeutic efficacy remains to be established, emerging evidence indicates that endovascular treatment of venous outflow obstruction can be performed safely in carefully selected patients. In a longitudinal cohort of patients with intracranial hypertension and venous stenosis, 32 underwent stenting of the internal jugular veins. During an average follow-up of 30 months, no major procedural complications were observed, and only isolated instances of restenosis and in-stent thrombosis were reported [24]. These findings provide preliminary evidence supporting the procedural safety of venous angioplasty and stenting as potential approaches to improve supracardiac venous outflow. Additionally, preliminary unpublished data from 20 patients in our STANDUP 1 registry similarly suggest that the procedure is well tolerated. Among these patients, one serious adverse event occurred, involving exacerbation of COPD requiring overnight hospital observation. No other procedure-related complications were observed. Following venous angioplasty or stenting, 14 patients demonstrated a statistically significant reduction in the magnitude of systolic blood pressure drop upon standing, with an average improvement of 14.6 mm Hg with 7 patients showing an improvement of more than 20 mm Hg.

Collectively, the proposed mechanisms and preliminary procedural data suggest that cerebral venous congestion resulting from venous outflow narrowing may contribute to autonomic and circulatory dysfunction in orthostatic patients. We hypothesize that improvement of venous outflow through supracardiac venous angioplasty or stenting may ameliorate cerebral venous congestion and could potentially mitigate symptoms of OH or OI, at least to a tolerable degree.

Objective

To evaluate the efficacy and safety of supracardiac venous angioplasty and stenting in improving orthostatic hypotension and orthostatic intolerance symptoms, alongside quality of life, in patients with jugular or brachiocephalic vein stenosis, using validated clinical tests.

Hypothesis

We hypothesize that supracardiac venous outflow obstruction contributes to orthostatic hypotension and orthostatic intolerance via elevated intracranial pressure, impaired glymphatic clearance, and disruption of vestibulo-sympathetic reflex integrity. Correcting venous stenosis via angioplasty and stenting will improve venous drainage, reduce ICP, restore glymphatic function, and enhance sympathetic anticipation, leading to clinically meaningful reductions in orthostatic blood pressure drop and symptom burden.

Methods

Intervention

The procedure follows standard sterile techniques and includes the following steps, with RHC integration:

- **Access:** Vascular access is achieved using sterile Seldinger technique. Common sites include common femoral veins, basilic veins, or internal/external jugular veins. Sheath sizes range from 5F to 9F. Arterial access for diagnostic angiography may be used if needed.
- **Arteriography:** Performed on bilateral common carotid, internal carotid, subclavian, and vertebral arteries with specified views (cervical, cranial, thoracic). Features such as stenosis, occlusion, collateral circulation are assessed, stagnation, reflux and venous phase anomalies are captured.
- **Venography:** Performed on bilateral internal jugular and subclavian veins with cervical, cranial, and thoracic views. (Includes detailed assessment of catheter position, reflux, time to contrast clearance, structural anomalies [e.g., thrombosis, aneurysm, diverticulum, atresia], collaterals [mild/moderate/severe], venous structural anomalies, flow direction in the superior ophthalmic vein [antegrade/retrograde], and visualization of the middle meningeal vein [yes/no]. Occipital sinus presence [yes/no] is also noted.)
- **Intravascular Ultrasound (IVUS):** Advanced into key venous structures (superior sagittal sinus, torcula, bilateral transverse/sigmoid sinuses, internal jugular veins, brachiocephalic veins). (Each vein is

segmented into three parts for structural data capture, including smoke, organized thrombus, extrinsic compression, and vascular compression.)

- **Right Heart Catheterization (RHC):** Integrated into the procedure to evaluate hemodynamic changes pre- and post-stenting. A Swan-Ganz catheter is advanced via venous access to measure:
 - Right atrial (RA) pressure
 - Right ventricular (RV) pressure (systolic/diastolic)
 - Pulmonary artery (PA) pressure (systolic/diastolic/mean)
 - Pulmonary capillary wedge pressure (PCWP)
 - Oxygen saturation (%)
 - Cardiac output (L/min) via thermodilution or Fick method
 - Cardiac index (L/min/m sq)
 - Internal jugular vein (IJV) temperature (°C)
 - Central Vein temperature (°C)

Measurements are taken before and immediately after stenting to assess improvements in cardiac output and repeated at 6 months to determine long term improvement. Waveforms and pressures are recorded for physiological analysis, including respiratory variation (poor/some/moderate/excellent) at relevant catheter stations.

- **Venous Angioplasty:** Performed based on IVUS/angiography. Balloons: Recommended 4 to 9 mm for intracranial stenoses; 10 to 14 mm for extracranial stenoses. Inflation: Recommended nominal for 10 seconds.
- **Venous Stenting:** Stents (Abre Stents are the only recommended ones). Sizing via angiography and IVUS. Post-stenting angioplasty with balloons (recommended armada or dorado). Follow-up imaging: Venogram and IVUS.
- **Hemostasis and Closure:** Manual compression or closure devices for access sites.

Facial Tone Assessment

Standardized anterior and oblique facial photographs will be obtained at baseline (pre-procedure) and post-procedure follow-up (day 1, 2 to 4 weeks, and 3 months).

- Images will be analyzed using automated facial landmark recognition software (e.g., MediaPipe FaceMesh or equivalent) to quantify changes in facial symmetry, contour, and periorbital fullness as exploratory markers of venous congestion resolution.
- Lighting, camera distance, and facial expression will be standardized.
- Images will be de-identified and stored securely in compliance with HIPAA and institutional data policies.

Facial Temperature Measurement (Infrared Thermometry)

Non-contact infrared temperature measurements will be obtained from the periorbital and lateral neck/IJV regions in both supine and standing positions to assess changes in venous congestion and superficial blood flow.

- Measurements will be performed at baseline and all scheduled post-procedure follow-ups (day 1, 2 to 4 weeks, 3 months).
- A handheld infrared thermal thermometer will be used, positioned approximately 3 to 5 cm from the

skin surface.

- Temperature differences between supine and standing positions, and pre- vs post-stenting, will be analyzed as exploratory indicators of venous outflow improvement.
- All data will be recorded under study ID and stored on secure, encrypted institutional servers.

Medication De-Escalation Protocol

Following intervention, this is the preferred sequence of medication discontinuation: droxidopa (Northera) pyridostigmine (Mestinon) fludrocortisone (Florinef) midodrine. On each day, if the patient is taking the next drug in sequence, stop it completely; if not, proceed to the next drug on the list. Continue orthostatic vitals (supine, seated, standing) and symptom checks (dizziness, presyncope, fatigue). If patient cannot tolerate the medication discontinuation, we will transition to a slower outpatient taper as clinically indicated.

Midodrine Equivalents (ME) Overview

Midodrine Equivalents (ME) provide a standardized way to compare the burden of different blood pressureraising medications. Each drugs usual maximum daily dose is defined as 100 ME, and lower doses are expressed as proportional values. A patients total ME is the sum of all agents taken.

Table 1: Midodrine Equivalents Conversion Table

Medication	Max Daily Dose	ME per mg	Common Doses ME
Midodrine	30 mg	3.33	10 mg→33 / 20 mg→67 / 30 mg→100
Fludrocortisone	0.3 mg	333.33	0.1 mg → 33 / 0.2 mg→67 / 0.3 mg→100
Droxidopa	1800 mg	0.056	600 mg→33 / 1200 mg→67 / 1800 mg→100
Pyridostigmine	180 mg	0.556	60 mg→33 / 120 mg→67 / 180 mg→100

Note: ME assumes proportional doseeffect within usual ranges; clinical response varies. Use clinical judgment for tapering and re-initiation.

Endpoints

Primary

1. Improvement in OHQ score post-intervention.
2. Improvement in orthostatic blood pressure drop (supine to standing) post-intervention.

Secondary

1. Reduction in total prescribed midodrine-equivalents post-intervention.
2. Improvement in QOL

Safety Monitoring

Post-Procedure: Vitals q15min x 2, then q30min x 1 (total 1 hour). Access site and neurological checks. Follow-ups: Recommended 24 weeks, ~3 months, ~6 months, ~1 year and ~2 years.

- Reporting Adverse Events (AEs): The management of adverse events (AEs) and serious adverse events (SAEs) during and after interventional procedures involves a structured, proactive approach to ensure

patient safety and mitigate risks. Once an AE is identified, the immediate priority is to stabilize the patient and address the complication based on its severity. For minor events, supportive measures such as observation or symptomatic treatment may suffice, while major complications may require urgent therapeutic interventions, such as additional procedures, escalation of care, or surgical consultation. The current escalation protocols are dictated by the cath lab and are critical and ensure that all team members are trained to recognize complications early and initiate appropriate responses. For example, in cases of bleeding or vascular injury, measures such as hemostatic agents, transfusions, or endovascular repair can be employed promptly.

Post-procedurally, patients are closely monitored for early signs of complications. This includes regular clinical assessments, imaging when necessary, and laboratory evaluations (e.g., coagulation profiles). Clear discharge instructions are provided for outpatient cases, detailing symptoms that may indicate complications (e.g., swelling, pain, fever) and emergency contact information for the care team. Serious AEs are documented and reported according to institutional protocols. Root cause analysis is conducted for SAEs to identify contributing factors whether technical, procedural, or systemic and develop strategies to prevent recurrence. Recommendations from these reviews may lead to updates in procedural protocols, enhanced training programs, and improved patient pathways to strengthen overall safety of the procedure.

Funding

- Research Assistant - Provided by Interventional Neuro Associates
- Mentoring - Provided by St. Francis Research Center
- Principal and co investigators will volunteer their time

Scientific Benefit

This study will provide insights into venous outflows role in OH/OI, including vestibulo-sympathetic mechanisms and venous tone. It evaluates RHC-guided interventions and medication de-escalation, potentially reducing pharmacological dependence and advancing VOODO (Venous Outflow Obstruction Disorders) understanding.

Subject Recruitment and Selection

Patients with OH/OI, recruited from neurology, primary care, cardiology. Aim: 200 participants over a 24-month period.

Location

St. Francis Hospital and Heart Center, 100 Port Washington Blvd, Roslyn, NY 11576.

Research Design

Prospective: Single-arm, unblinded registry trial.

- Experimental Cohort:
 - Inclusion Criteria:
 1. Age 18 years or older
 2. Diagnosed OH as defined above, or OI defined by a score greater than 3 on the orthostatic hypotension questionnaire (OHQ)
 3. Patient/surrogate/healthcare proxy able to provide informed consent which will be obtained prior to the procedure

4. Severe venous stenosis in the supraceliac venous outflow tract draining the head and neck including the transverse and sigmoid sinuses, internal jugular veins and brachiocephalic veins.

Venous stenosis will be classified as severe based on the presence of one or more of the below criteria

a) Radiographic stenosis: $>50\%$ reduction in lumen diameter compared to the adjacent normal segment (or expected normal diameter based on patient anatomy) PLUS one of

- * Contrast Stagnation greater than 10 seconds (Normal < 4 seconds)
- * Reflux into collaterals
- * Varicosities in or around the valves
- * Filling defects (thrombus / organized thrombus / septae / webs)

b) IVUS based stenosis: $>75\%$ reduction in areas compared to the adjacent normal segment PLUS one of

- * Smoke (swirling mixed echogenic signals)
- * Increased speckling density
- * Valve abnormalities (thickened valve, poor mobility)
- * Thrombus
- * Recanalized thrombus

c) Pressure Gradient: ≥ 3 mm Hg pressure gradient

d) Pressure Waveform: Damped or monophasic waveform

– Exclusion Criteria:

- * Pregnancy/breastfeeding
- * Active infection
- * Coagulopathy or contraindication to venous stenting
- * Stenosis at level of C1
- * Prior radiation to head, neck or body
- * Pacemaker ports or leads
- * Patients undergoing hemodialysis
- * Baseline MRS 5 or greater
- * Poor mobility at baseline, bedridden

- Control Cohort:

– Inclusion Criteria:

1. Age 18 years or older
2. Undergoing IJV angioplasty and/or stenting for non-orthostatic indications (e.g., idiopathic intracranial hypertension, venous outflow obstruction, intracranial hypertension without orthostatic symptoms)

3. Orthostatic hypotension not present, defined by: a. Absence of symptomatic OI, and b. Normal orthostatic vital signs (no sustained SBP 20 mmHg or greater and DBP 10 mmHg or greater upon standing)
 4. Patient/surrogate/healthcare proxy able to provide informed consent which will be obtained prior to the procedure
- Exclusion Criteria:
- * Pregnancy/breastfeeding
 - * Active infection
 - * Coagulopathy or contraindication to venous stenting
 - * Stenosis at level of C1
 - * Prior radiation to head, neck or body
 - * Pacemaker ports or leads
 - * Patients undergoing hemodialysis
 - * Baseline MRS 5 or greater
 - * Poor mobility at baseline, bedridden

Statistical Analysis

- Paired t-tests for pre/post measures; repeated measures ANOVA for QoL. Sample size: 200 for 97.8% power ($p < 0.05$).

Potential Risks

- Intracranial Hemorrhage (Bleeding in the Brain)
- Intrapinal Hemorrhage (Bleeding in the Spinal Canal)
- Stroke
- Venous Infarction (Brain Tissue Damage from Blocked Veins)
- Renal Failure (Kidney Damage)
- Contrast Allergy
- Groin Hematoma (Bleeding at the Access Site)
- Pseudoaneurysm (False Aneurysm at Access Site)
- Permanent or Long-Term Post-Procedural Pain at Intervention Sites
- Mortality (Death) Other Possible Risks:
 - Infection
 - Radiation Exposure
 - Thrombosis

Potential Benefits

- Symptom Improvement
- Better Quality of Life
- Contribution to research and further understanding of this condition

Protection of Subjects

- **De-identification & Data Security:** All data are de-identified at collection using a unique study ID; the linkage file is stored separately. Data are stored on encrypted servers (encryption at rest and in transit), with role-based access and audit logs. Retention is 5 years, then purge/archival per institutional policy.
- **Minimal Necessary PHI:** Only the minimum PHI required for study conduct is accessed; exports for analysis use de-identified datasets.
- **Informed Consent Process:** Consent obtained by trained study staff with ample time for questions/answers; teach-back is used to confirm understanding; language-appropriate forms and interpreter services are offered; participation is voluntary and can be withdrawn at any time without impact on clinical care.
- **DSMB Oversight:** The team will review safety/efficacy data quarterly or per every 30 subjects enrolled, whichever comes first; convene ad hoc meeting with the St. Francis Research stewardship for serious/unanticipated problems; may recommend protocol modification, pause, or termination.
- **Adverse Event Monitoring & Reporting:** Bedside clinical monitoring plus protocol AE/SAE capture with prompt reporting to the IRB/DSMB per institutional timelines; clear pathways for medical management and follow-up.
- **Risk Minimization:** Standardized procedural techniques, lowest reasonable contrast/radiation, renal protection when indicated, and immediate escalation pathways for complications.
- **Confidentiality & Sharing:** Any data sharing uses de-identified datasets; external sharing requires IRB/DSMB approval and data-use agreements.

Schedule of Assessments for STANDUP 2 Study

Assessment/Procedure	Screening	Baseline	Procedure Day	Day 1 (Post)	4 Week (±14 d)	3 M (±20 d)	6 M (±60 d)	12 M (±60 d)	Unsch
Informed Consent	✓								
Medical History/Physical Exam	✓	✓	✓ (pre/post)	✓	✓	✓	✓	✓	✓
Inclusion/Exclusion Criteria Review	✓	✓							
Vital Signs (BP, HR, etc., including orthostatic)	✓	✓	✓ (intra-op)	✓ (q15min x2, q30min x1)	✓	✓	✓	✓	✓ (if ind.)
Laboratory Tests (CBC, CMP, PT/PTT, Pregnancy Test if applicable)	✓	✓							✓ (if ind.)

Assessment/Procedure	Screening	Baseline	Procedure Day	Day 1 (Post)	2-4 Weeks (±7 d)	3 Months (±14 d)	6 Months (±30 d)	12 Months (±30 d)	Un-scheduled
Diagnostic Imaging/Hemodynamics			✓			✓ (if ind. for patency)	✓ (if ind.)	✓ (if ind.)	✓ (if ind.)
- Venography (radiographic stenosis >50%, delayed clearance >10 sec, reflux, stagnation)			✓			✓ (if ind.)	✓ (if ind.)	✓ (if ind.)	✓ (if ind.)
- IVUS (CSA >50%, swirling/smoke, speckling/stagnation, poor valve mobility, recanalized thrombus)			✓						✓ (if ind.)
- Pressure Gradients/Waveforms (≥3 mmHg gradient, damped/monophasic, stagnant flow)			✓			✓ (if ind.)	✓ (if ind.)	✓ (if ind.)	✓ (if ind.)
Intervention (Angioplasty/Stenting if qualified)			✓						
Facial Images (for venous congestion assessment)		✓		✓	✓	✓	✓	✓	✓ (if ind.)
Facial Temperature Measurement (Infrared Thermometry)		✓		✓	✓	✓	✓	✓	✓ (if ind.)
Symptom Questionnaires		✓		✓	✓	✓	✓	✓	✓ (if ind.)
- OHQ (Orthostatic Hypotension Questionnaire)		✓		✓	✓	✓	✓	✓	✓ (if ind.)
Quality of Life (SF-36 or equivalent)		✓		✓	✓	✓	✓	✓	✓ (if ind.)
Adverse Events/Safety Monitoring		✓	✓	✓	✓	✓	✓	✓	✓
- Complications (e.g., thrombosis, perforation, bleeding; Clavien-Dindo classification)			✓	✓	✓	✓	✓	✓	✓

Assessment/Procedure	Screening	Baseline	Procedure Day	Day 1 (Post)	2-4 Weeks (≤7 d)	3 Months (≤14 d)	6 Months (≤30 d)	12 Months (≤30 d)	Un-scheduled
Concomitant Medications Review (for Midodrine Equivalents)	✓	✓	✓	✓	✓	✓	✓	✓	✓
Study Exit/Discontinuation								✓ (or early)	✓ (if drop-out)

References

1. Bradbury S, Eggleston C. Postural hypotension: A report of three cases. American heart journal [Internet]. 1925 Oct 1 [cited 2024 Dec 26];1(1):7386. Available from: <https://www.sciencedirect.com/science/article/pii/S0002870325900075>
2. Rutan GH, Hermanson B, Bild DE, Kittner SJ, LaBaw F, Tell GS. Orthostatic hypotension in older adults. The Cardiovascular Health Study. CHS Collaborative Research Group. Hypertension [Internet]. 1992 Jun;19:50819. Available from: <https://www.ncbi.nlm.nih.gov/pubmed/1592445>
3. Freeman R, Wieling W, Axelrod FB, Benditt DG, Benarroch E, Biaggioni I, et al. Consensus statement on the definition of orthostatic hypotension, neurally mediated syncope and the postural tachycardia syndrome. Clin auton res [Internet]. 2011 Apr;21(2):6972. Available from: <https://www.ncbi.nlm.nih.gov/pubmed/21431947>
4. Ricci F, Fedorowski A, Radico F, Romanello M, Tatasciore A, Di Nicola M, et al. Cardiovascular morbidity and mortality related to orthostatic hypotension: A meta-analysis of prospective observational studies. Eur heart j [Internet]. 2015 Jul 1;36(25):160917. Available from: <https://www.ncbi.nlm.nih.gov/pubmed/25852216>
5. Ko D, Preis SR, Lubitz SA, McManus DD, Vasan RS, Hamburg NM, et al. Relation of Orthostatic Hypotension With New-Onset Atrial Fibrillation (From the Framingham Heart Study). The american journal of cardiology [Internet]. 2018 Mar 1 [cited 2023 Jul 18];121(5):596601. Available from: <https://www.sciencedirect.com/science/article/pii/S0002914917318611>
6. Bell EJ, Agarwal SK, Cushman M, Heckbert SR, Lutsey PL, Folsom AR. Orthostatic Hypotension and Risk of Venous Thromboembolism in 2 Cohort Studies. Am j hypertens [Internet]. 2016 May [cited 2023 Jul 11];29(5):63440. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5014082/>
7. Franceschini N, Rose K, Astor BC, Couper D, Vupputuri S. Orthostatic hypotension is associated with incident chronic kidney disease: The Atherosclerosis Risk In Communities Study. Hypertension [Internet]. 2010 Dec [cited 2023 Jul 26];56(6):10549. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3147024/>
8. Kulkarni S, Jenkins D, Dhar A, Mir F. Treating Lows: Management of Orthostatic Hypotension. J cardiovasc pharmacol [Internet]. 2024 Sep 3 [cited 2025 Jan 1];84(3):30315. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11368167/>
9. Sra J, Maglio C, Biehl M, Dhala A, Blanck Z, Deshpande S, et al. Efficacy of Midodrine Hydrochloride in Neurocardiogenic Syncope Refractory to Standard Therapy. Journal of cardiovascular electrophysiology [Internet]. 1997 [cited 2023 Jul 18];8(1):426. Available from: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1540-8167.1997.tb00607.x>

10. Vaidyanathan S, Soni BM, Hughes PL. Midodrine: Insidious development of urologic adverse effects in patients with spinal cord injury: A report of 2 cases. *Adv ther* [Internet]. 2007;24(4):71220. Available from: <https://www.ncbi.nlm.nih.gov/pubmed/17901021>
11. Burns A, Brown TM, Semple P. Extreme metabolic alkalosis with fludrocortisone therapy. *Postgrad med j* [Internet]. 1983 Aug [cited 2025 Jan 6];59(694):5067. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2417584/>
12. Willis FR, Byrne GC, Jones TW. Fludrocortisone induced heart failure in Addisons disease. *J paediatr child health* [Internet]. 1994 Jun;30(3):2801. Available from: <https://www.ncbi.nlm.nih.gov/pubmed/8074919>
13. Kaufmann H, Freeman R, Biaggioni I, Low P, Pedder S, Hewitt LA, et al. Droxidopa for neurogenic orthostatic hypotension: A randomized, placebo-controlled, phase 3 trial. *Neurology* [Internet]. 2014 Jul 22 [cited 2024 Nov 12];83(4):328. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC4115605/>
14. Remijn-Nelissen L, Verschuuren JJGM, Tannemaat MR. The effectiveness and side effects of pyridostigmine in the treatment of myasthenia gravis: A cross-sectional study. *Neuromuscular disorders* [Internet]. 2022 Oct 1 [cited 2024 Nov 12];32(10):7909. Available from: [https://www.nmd-journal.com/article/S0960-8966\(22\)00652-6/fulltext](https://www.nmd-journal.com/article/S0960-8966(22)00652-6/fulltext)
15. Levick JR. *An introduction to cardiovascular physiology*. London ; Butterworths; 1991.
16. Magder S. The meaning of blood pressure. *Crit care* [Internet]. 2018 Oct 11 [cited 2025 Jan 1];22:257. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6180453/>
17. Shapiro E. Adolf Fick forgotten genius of cardiology. *Am j cardiol* [Internet]. 1972 Nov 8;30(6):6625. Available from: <https://www.ncbi.nlm.nih.gov/pubmed/4563215>
18. Aoki M, Sakaida Y, Tanaka K, Mizuta K, Ito Y. Evidence for vestibular dysfunction in orthostatic hypotension. *Exp brain res* [Internet]. 2012 Mar 1 [cited 2025 Sep 5];217(2):2519. Available from: <https://doi.org/10.1007/s00221-011-2989-0>
19. Normahani P, Shalhoub J, Narayanan S. Repurposing the systemic venous return model for conceptualisation of chronic venous insufficiency and its management. *Phlebology* [Internet]. 2020 Dec [cited 2025 Jan 1];35(10):74951. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7675772/>
20. Wang Y, Payne SJ. Static autoregulation in humans. *J cereb blood flow metab* [Internet]. 2024 Nov [cited 2025 Jan 1];44(11):1191207. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11542139/>
21. Hallgren E, Migeotte PF, Kornilova L, Delière Q, Fransen E, Glukhikh D, et al. *Dysfunctional vestibular system causes a blood pressure drop in astronauts returning from space*. *Scientific reports*. 2015;5:17627.
22. Yates BJ, Miller AD, Perlmutter SI. Plastic changes in processing of graviceptive signals during spaceflight potentially contribute to postflight orthostatic intolerance. *Journal of vestibular research*. 2003;13(4-6):395404.
23. Ketch T, Biaggioni I, Robertson R, Robertson D. Four faces of baroreflex failure: Hypertensive crisis, volatile hypertension, orthostatic tachycardia, and malignant vagotonia. *Circulation* [Internet]. 2002 May 28;105(21):251823. Available from: <https://www.ncbi.nlm.nih.gov/pubmed/12034659>
24. Bai C, Chen Z, Ding Y, Ji X, Yuan J, Meng R. Longterm safety and efficacy of stenting on correcting internal jugular vein and cerebral venous sinus stenosis. *Ann clin transl neurol* [Internet]. 2023 Jun 5 [cited 2025 Nov 23];10(8):130513. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10424652/>

